Hints and Tips for OASP Assignment 1

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Exercise 1: Basic Probability Concepts

(a) Finding Probabilities for Events:

- Identify sample space and list all possible outcomes.
- Use basic probability formula $P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$.

(b) Conditional Probability:

- Use $P(A|B) = \frac{P(A \cap B)}{P(B)}$, assuming $P(B) > 0$.
- Define intersections of events for common outcomes.

(c) Independence of Events:

- Events A and B are independent if $P(A \cap B) = P(A) \cdot P(B)$.

Exercise 2: Expectation and Variance

(a) Expected Value (Mean):

- For discrete X, $E[X] = \sum_x x \cdot P(X = x)$; for continuous X, $E[X] = \int x f_X(x)\, dx$.

(b) Variance:

- Use $\text{Var}(X) = E[X^2] - (E[X])^2$.
- For $E[X^2]$, sum (or integrate) over $x^2 \cdot P(X = x)$ for discrete, $\int x^2 f_X(x)\, dx$ for continuous.

(c) Standard Deviation:

- $\sigma_X = \sqrt{\text{Var}(X)}$, the square root of the variance.

Exercise 3: Auto-covariance Matrix and Correlation Coefficient

(a) Marginal PMFs p_{X_1} and p_{X_2}:

- Use $p_{X_1}(x_1) = \sum_{x_2} p_{X_1,X_2}(x_1, x_2)$ and $p_{X_2}(x_2) = \sum_{x_1} p_{X_1,X_2}(x_1, x_2)$.

(b) Conditional PMF $p_{X_2|X_1}$:

- $p_{X_2|X_1}(x_2|x_1) = \frac{p_{X_1,X_2}(x_1,x_2)}{p_{X_1}(x_1)}$.

(c) Mean Vector μ_X:

- $\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \end{bmatrix}$ where $E[X_1] = \sum_{x_1} x_1 \cdot p_{X_1}(x_1)$ and similarly for $E[X_2]$.

(d) Auto-covariance Matrix C_{XX}:

- $C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{bmatrix}$.
- Calculate $\mathrm{Var}(X_i) = E[X_i^2] - (E[X_i])^2$ and $\mathrm{Cov}(X_1, X_2) = E[X_1X_2] - E[X_1]E[X_2]$.

(e) Correlation Coefficient:

- $\rho_{X_1,X_2} = \frac{\mathrm{Cov}(X_1,X_2)}{\sqrt{\mathrm{Var}(X_1)\cdot\mathrm{Var}(X_2)}}$.

Exercise 4: Covariance and Correlation of Linearly Dependent Variables

(a) Covariance $\mathbf{Cov}(X, Y)$:

- For $Y = aX + b$, $\mathrm{Cov}(X, Y) = a \cdot \mathrm{Var}(X)$.

(b) Correlation Coefficient $\rho_{X,Y}$:

- $\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\cdot\mathrm{Var}(Y)}}$.
- Simplifies to $\frac{a}{|a|}$, resulting in ± 1 based on a's sign.

Exercise 5: Simulation and Estimation of Joint PMFs

(a) Generate Realizations:

- Map random numbers to joint PMF outcomes using cumulative probabilities.

(b) Estimate Joint PMF $\hat{p}_{X,Y}$:

- Count occurrences of each (X, Y) pair and normalize by total samples M.

(c) Estimate Marginal PMFs:

- Sum estimated joint PMF across rows for X and columns for Y.

(d) Estimate Correlation Coefficient $\hat{\rho}_{X,Y}$:

- $\hat{\rho}_{X,Y} = \frac{\frac{1}{M}\sum_{m=1}^{M} x_m y_m - \bar{x}\bar{y}}{\sqrt{\left(\frac{1}{M}\sum_{m=1}^{M} x_m^2 - \bar{x}^2\right)\left(\frac{1}{M}\sum_{m=1}^{M} y_m^2 - \bar{y}^2\right)}}$.

Exercise 6: Continuous Uniform Distribution and Transformation

(a) Deriving the PDF of $Y = X_1 + X_2$:

- Use convolution: $f_Y(y) = \int_0^1 f_{X_1}(x) f_{X_2}(y - x)\, dx$.
- Split into cases: $0 \leq y \leq 1$ and $1 < y \leq 2$.

(b) Calculating Mean and Variance of Y:

- $E[Y] = \int_0^2 y f_Y(y)\, dy$ and $E[Y^2] = \int_0^2 y^2 f_Y(y)\, dy$.
- Use $\text{Var}(Y) = E[Y^2] - (E[Y])^2$.

(c) Simulations and Histogram Plot:

- Simulate $Y = X_1 + X_2$ using samples of X_1 and X_2.
- Plot histogram of Y and overlay theoretical PDF.